Parallel circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate branch currents to a shared potential difference.

SUCCESS CRITERIA - I can...

1. Identify branches connected across the same two nodes.
2. Use equal voltage across parallel branches.
3. Add branch currents to find supply current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Electrons in a wire drift at **0.3 mm/s** while a lamp responds quickly after a switch closes. **Explain why response time cannot** be found by timing one electron across the wire.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\mathbf{1}/R_{total} = \mathbf{1}/R_1 + \mathbf{1}/R_2$; $I_{total} = I_1 + I_2$; $V_1 = V_2$. Parallel branches connect to the same two nodes and share voltage.

Worked reasoning: Two parallel branches of **6** and **3 ohms** each have **6 V** across them. Branch currents are **1** and **2 A**; source current is **3 A**.

Your turn: Draw the two branches and **mark where the currents split** and recombine.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark the two shared nodes in the parallel diagram. If one branch opens, can the other remain powered? Explain.



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A **12 V** supply powers **6 Ω** and **12 Ω** resistors in parallel.
1. **Label the shared nodes**.
2. **Calculate branch currents**, source current and equivalent resistance; check that the branch currents add to the source current.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

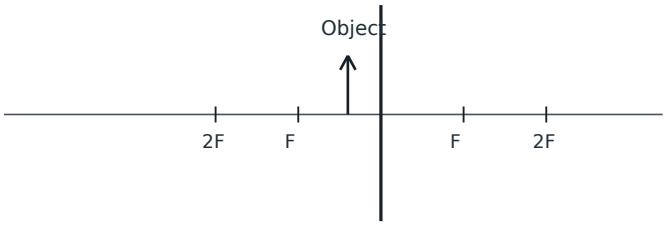
05 Explain your reasoning

Why can the equivalent resistance be smaller than either branch resistance?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A converging lens has f=**20 cm** and d_o=**8 cm**. Use **1**/f=**1**/d_o **+1**/d_i and m=-d_i/d_o. Find d_i, magnification and image kind.
Draw the rays used in your explanation.



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

07 Exit check

Two equal **8 Ω** resistors are in parallel.
1. **Find equivalent resistance**.
2. Compare with the same resistors in series.
3. **Label the circuit** and **show the current path or** paths.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **12 V** supply feeds parallel **4** and **6 ohm** resistors.
1. **Find equivalent resistance**, total current and total power by two methods.
2. **Label the circuit** and **show the current path or** paths.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③